

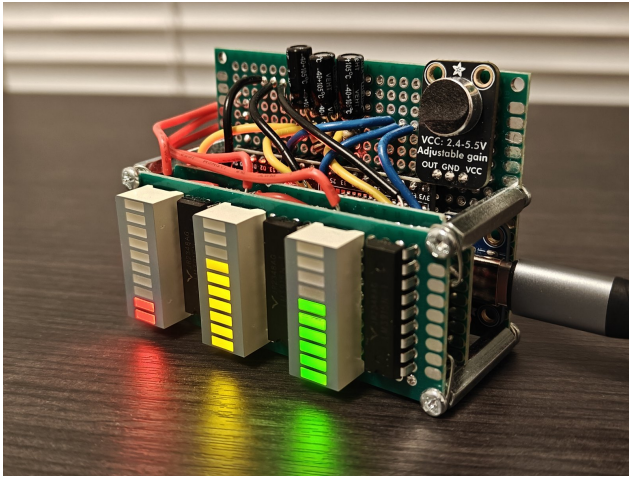
Audio Spectrum Visualizer

A desk light display that shows the bass, mids and treble of the music in the room as three LED bars

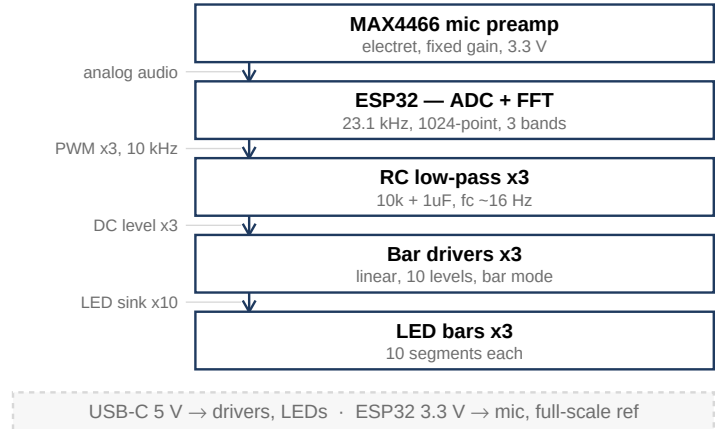
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Full documentation



SYSTEM ARCHITECTURE



WHAT IT DOES

A desk light display for anyone who likes to see their music. A microphone listens to the room; an ESP32 microcontroller splits the sound into bass, mids and treble with a Fast Fourier Transform and drives three 10-LED bars about 20 times a second. Each bar scales itself to the room's noise and the music's recent peaks, so it needs no calibration. Built for looks, not measurement.

HOW IT WORKS

- The ESP32's ADC samples at a measured 23.1 kHz; a 1,024-point FFT splits each block into 80-300 Hz, 300 Hz-4 kHz and 4-10 kHz.
- Each band's level in dB sets a PWM duty, and a 10k + 1uF filter turns the 10 kHz PWM into a steady voltage for its bar.
- Linear bar-graph driver chips light 10 LEDs per bar; because the input is proportional to dB, every LED is an equal dB step.

MEASURED RESULTS

Driver LED steps, 3 chips (multimeter)	0.33 V, linear
Sample rate / display rate (firmware timer)	23.1 kHz / 19.6 fps
Supply current at 5 V, dark / loudest seen	71 / 238 mA
ESP32 ADC linear range at 11 dB (multimeter)	0.24-2.65 V

KEY DECISIONS

- ESP32 over Uno or ESP32-C3: an FPU for the FFT.
- Fixed-gain mic over AGC: loud against quiet stays visible.
- No anti-alias filter: 13-19 kHz hi-hats fold into treble.
- PWM + RC filter as the DAC: 3 bands, only 2 on-chip DACs.

MEASURED EVIDENCE

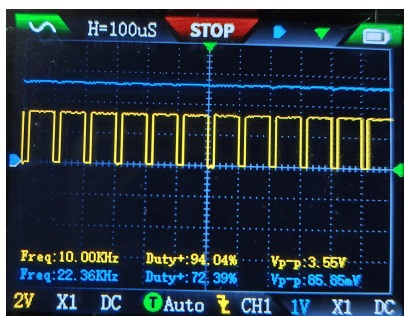


Fig. 1 - FNIRSI 2C53T, 0.1 ms/div, CH1 2 V/div, CH2 1 V/div, DC, x1

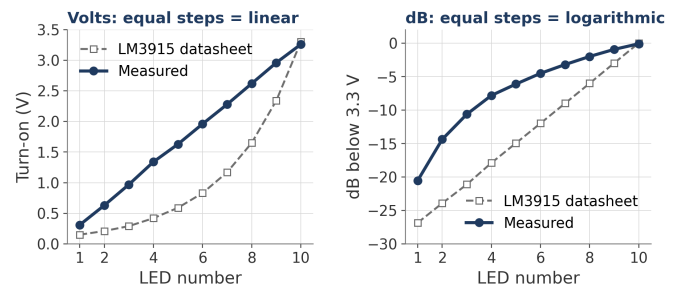


Fig. 2 - FNIRSI 2C53T DC V at pin 5, trimmer-driven, 3 chips